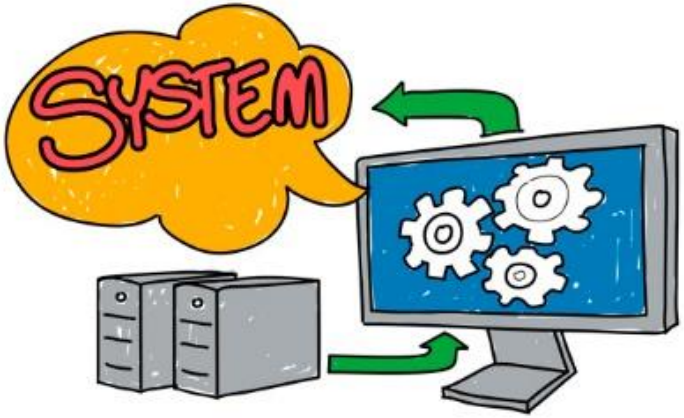


Introduction

BIS605

Software Development Technologies for Digital Business

Asst. Prof. Dr. Worarat Krathu

Course Outline

Lecture	Topic
1	- Introduction to System architecture and WWW
2	- Software Development Lifecycle and Web development process
3-4	- System design - Principle of web design - Design lab with Figma
5-7	- Software Technologies (Software frameworks, Github, Docker, Web services, IoT etc.)
8	Midterm Exam (25%)
9-13	- Web Technologies (HTML, CSS, JS, PhP, etc.)
14	- Mobile Technologies
15	- Lab quiz (20%)
16	- System Design Project presentation(20%)
17	Final Exam (25%)

References

- B. Cameron, E. Crawey, D. Selva, Systems Architecture, Global Edition, Peason, 2015
- https://msu.edu/course/lbs/126/lectures/Internet-history.ppt
- https://www.ntu.edu.sg/home/ehchua/programming/webprogramming/HTTP_Basics.html
- https://developer.mozilla.org/en-US/docs/Learn/Server-side/First_steps/Introduction
- https://en.wikipedia.org/wiki/Systems_architecture
- https://towardsdatascience.com/10-common-software-architectural-patterns-in-a-nutshell-a0b47a1e9013
- https://theappsolutions.com/blog/development/project-tech-stack/
- Lectures of:
  - Asst. Prof. Dr. Chonlameth Arpnikanondt
  - Asst. Prof. Sumet Angkasirikul
  - Ajarn Pichet Limvachiranan

SYSTEM ARCHITECTURE

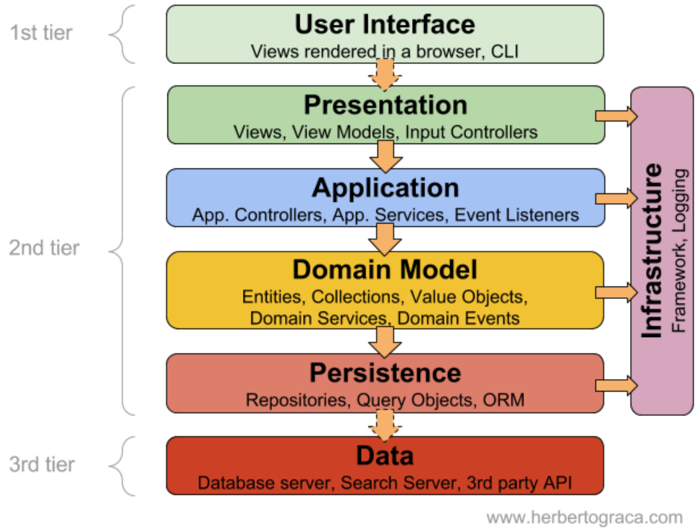
System Architecture

- A system is a group of interacting or interrelated entities that form a unified whole.
- In computer science, a system is a hardware system, software system, or combination, which has components as its structure and observable inter-process communications as its behavior.

- A system architecture is the conceptual model that defines the structure, behavior, and more views of a system. It is an abstract description of the entities of a system and the relationship between those entities.

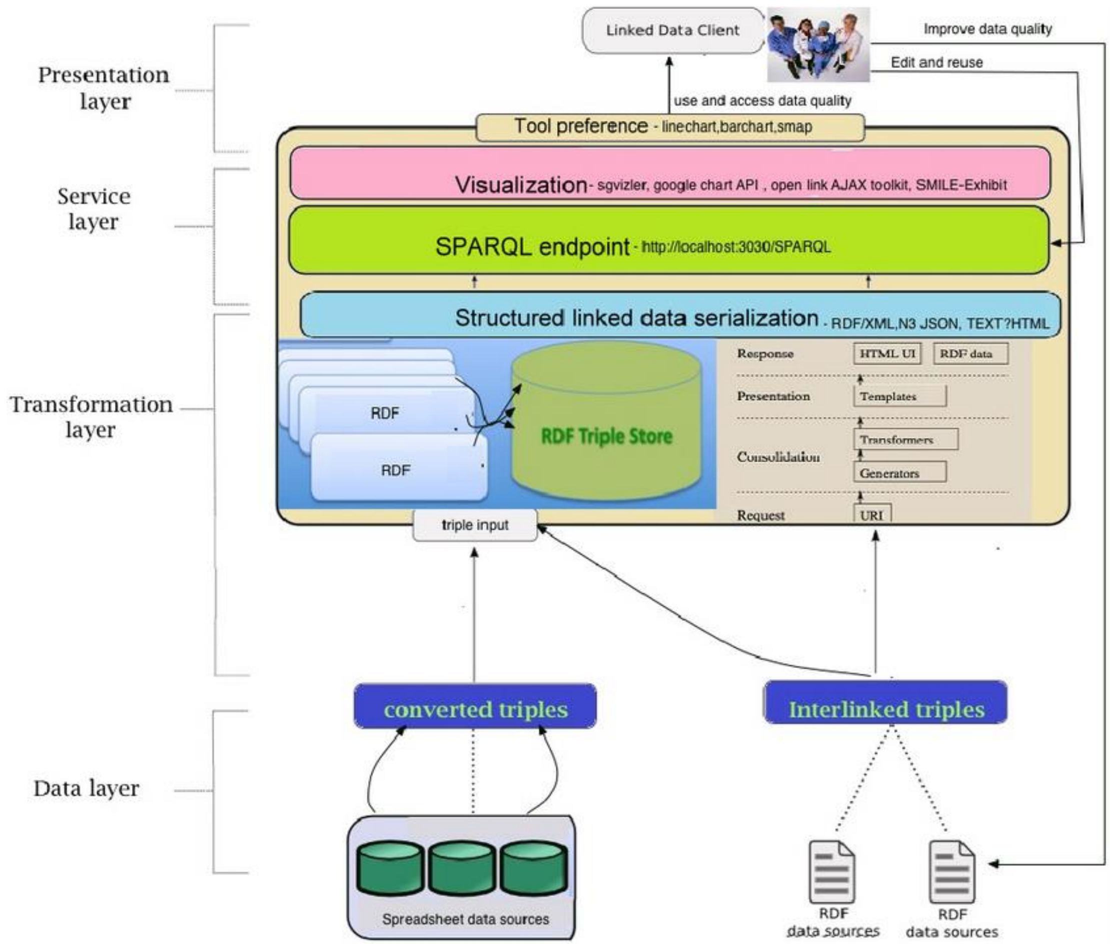
Common System Architecture Pattern

- Layered pattern
  - This pattern can be used to structure programs that can be decomposed into groups of subtasks, each of which is at a particular level of abstraction.
  - Each layer provides services to the next higher layer.
  - 4 common layers
    - Presentation layer (aka UI layer)
    - Application layer (aka service layer)
    - Business logic layer (aka domain layer)
    - Data access layer (aka persistence layer)



Common System Architecture Pattern

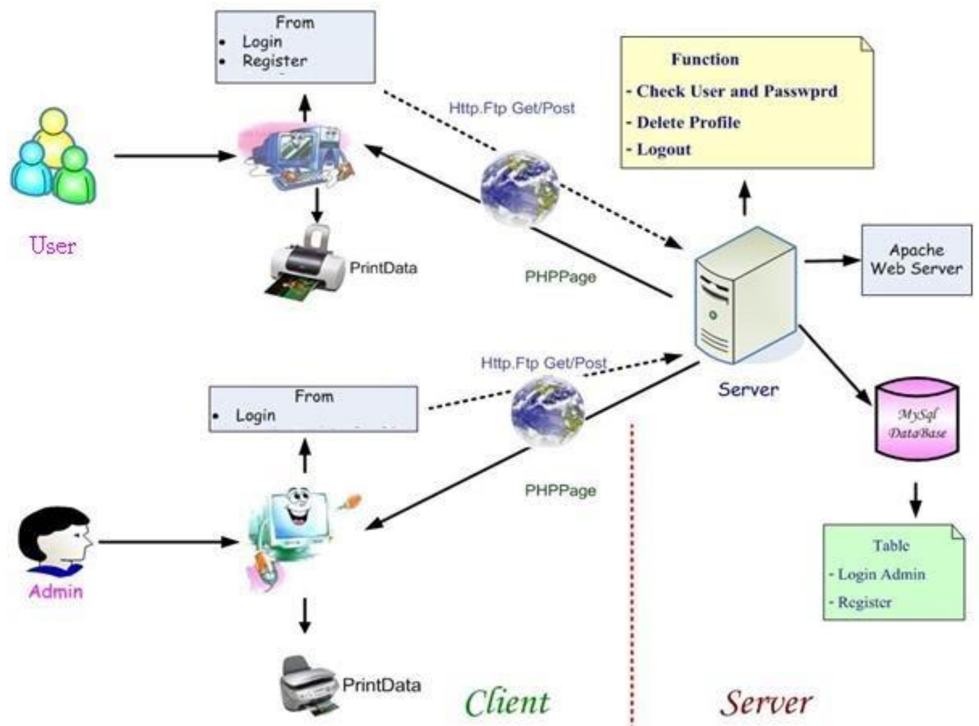
- Layered pattern



Credit: https://www.researchgate.net/figure/LOHD-System-architecture-with-all-the-four-layers-from-the-data-representation-layer-to_fig2_267777242

Common System Architecture Pattern

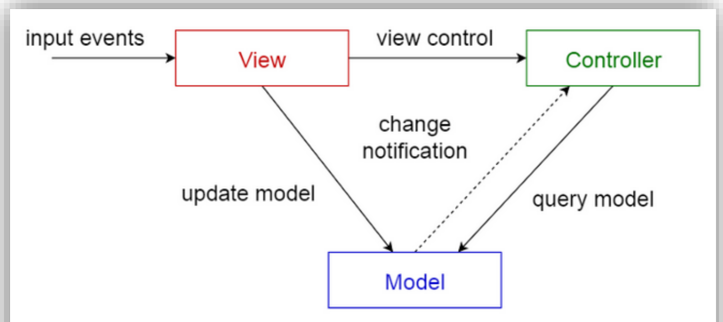
- Client-server pattern
  - This pattern consists of two parties; a server and multiple clients.



Credit: http://supapatcha360.blogspot.com/2009/02/system-architecture-design.html

Common System Architecture Pattern

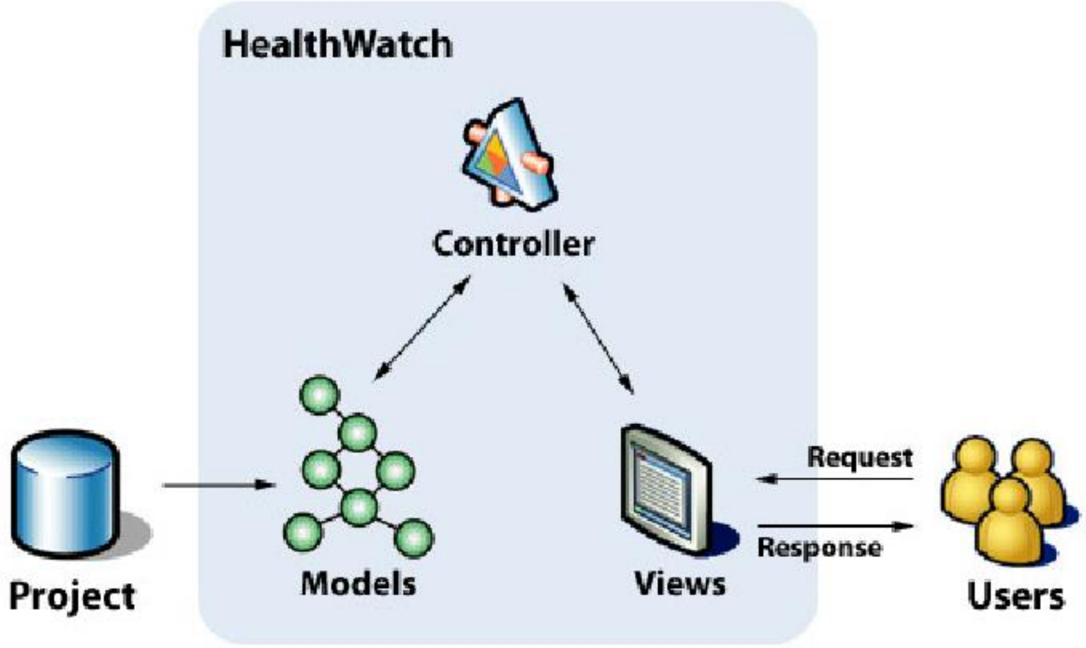
- Model-view-controller pattern
- This pattern, also known as MVC pattern, divides an interactive application in to 3 parts as,
  - **model** — contains the core functionality and data
  - **view** — displays the information to the user (more than one view may be defined)
  - **controller** — handles the input from the user



Model-view-controller pattern

Common System Architecture Pattern

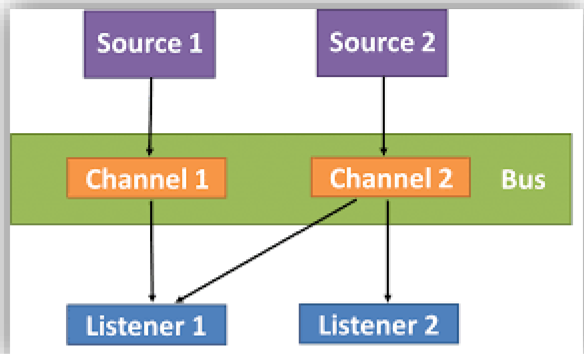
- Model-view-controller pattern

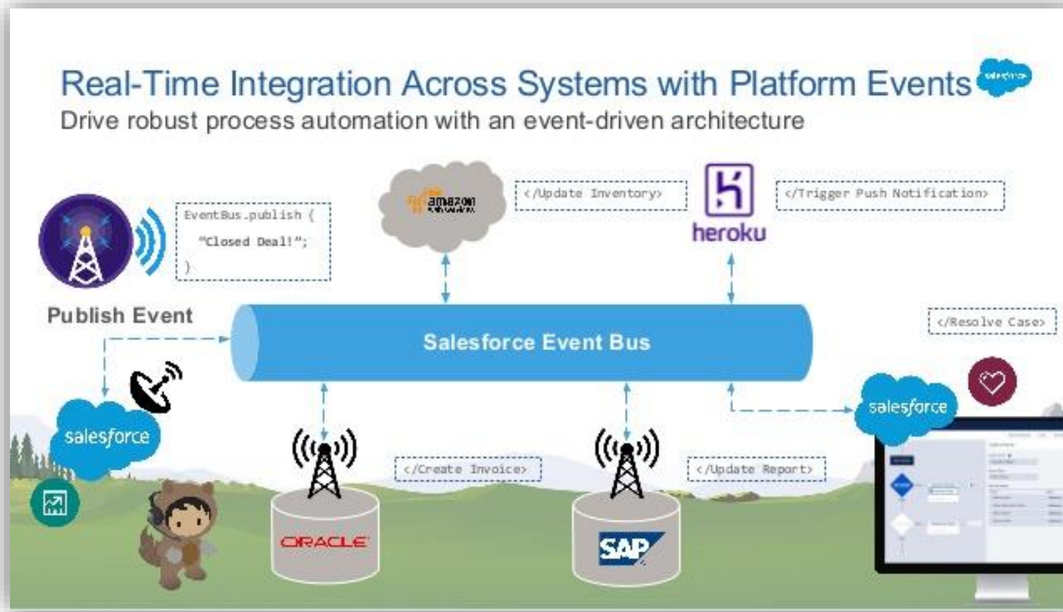


Credit: https://www.researchgate.net/figure/Proposed-System-Architecture-Using-the-MVC-Architecture_fig8_237726575

Common System Architecture Pattern

- Event-bus pattern
  - This pattern primarily deals with events and has 4 major components; event source, event listener, channel and event bus.

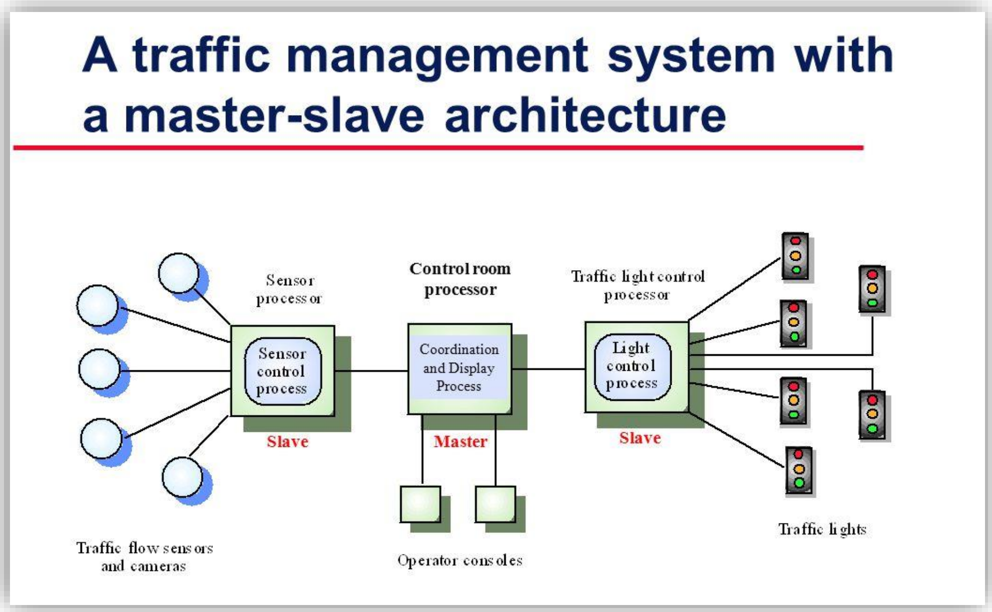




Credit: https://riccentre.ca/software-architecture-design-patterns/
https://www.slideshare.net/developerforce/replicate-salesforce-data-in-real-time-with-change-data-capture

Common System Architecture Pattern

- Master-slave pattern
  - This pattern focuses on two parties; master and slaves. The master component distributes the work among identical slave components, and computes a final result from the results which the slaves return.



Credit: https://slideplayer.com/slide/5923690/

Technology Stack

- Technology stack or "Tech Stack" is a set of tools developers use to make an application.

- It consists of software applications, frameworks, and programming languages that realize some aspects of the program.

- Structure-wise, the tech stack consists of two equal elements. One is the front-end or client-side; the other is the server-side or back end.

Technology Stack

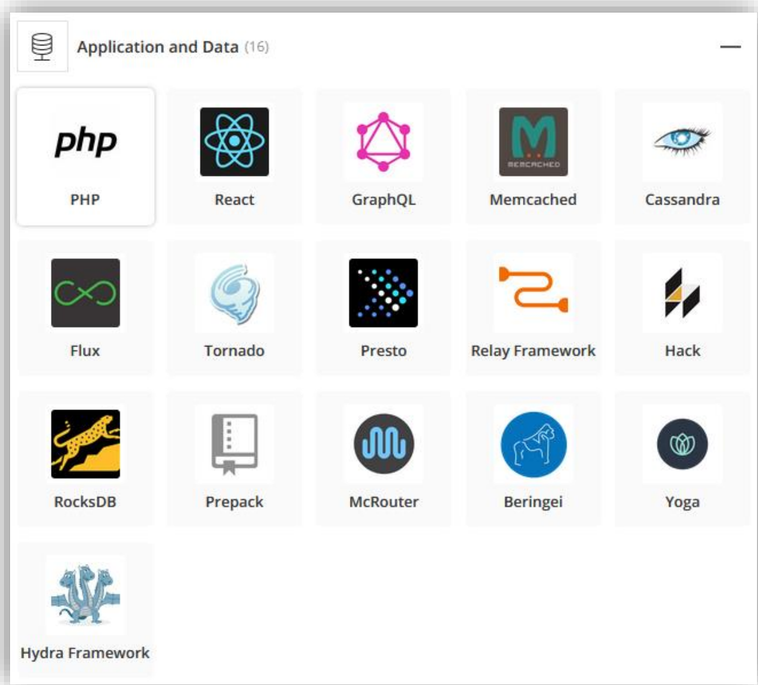
Credit: https://deepscope.com/th/deepscopes-tech-stack/

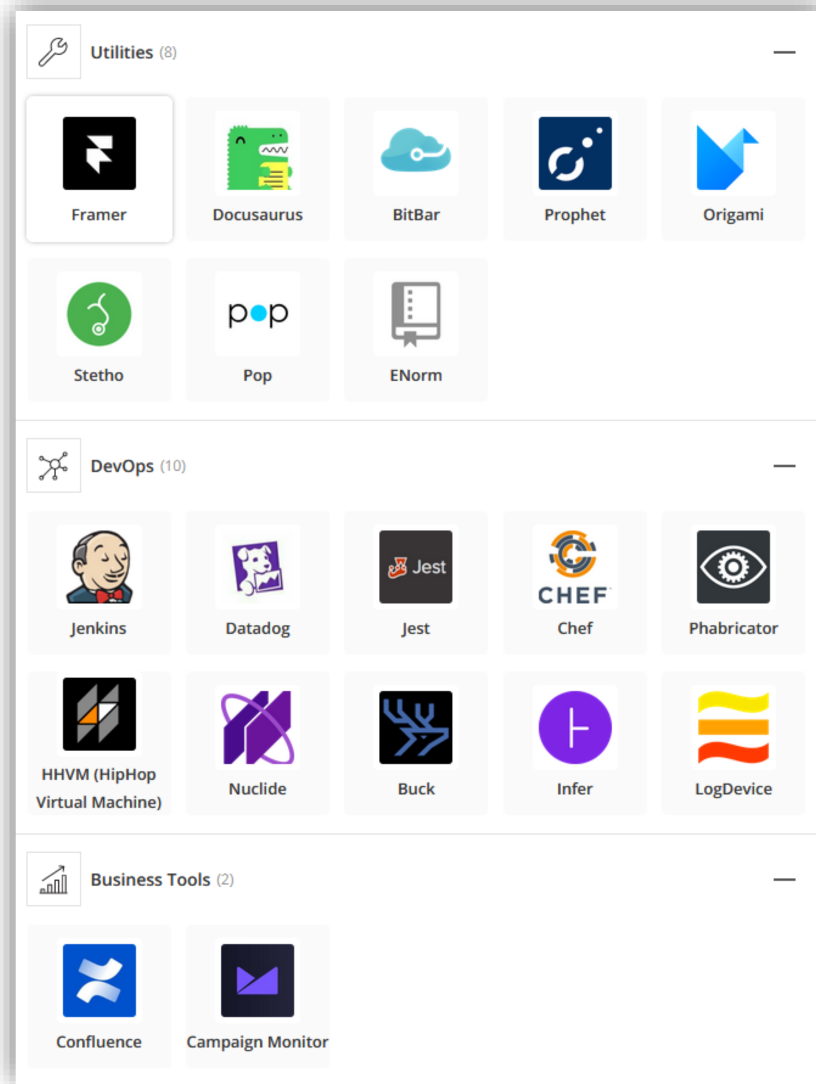
Technology Stack

Credit: https://ebaytech.berlin/picking-technology-stacks-2c4e138c12c8

Technology Stack

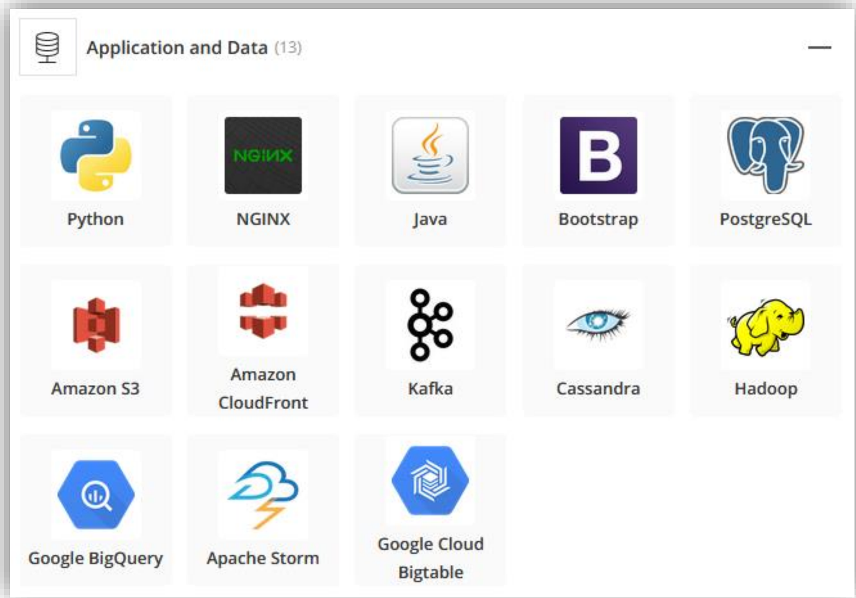
- Facebook

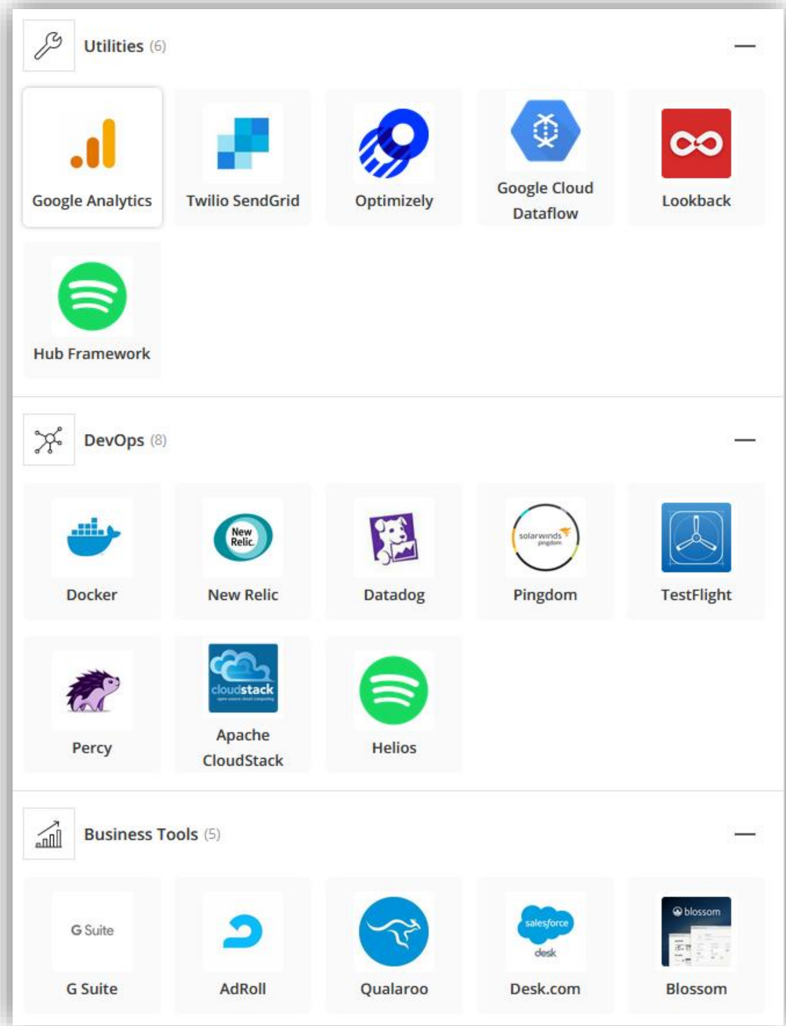




Technology Stack

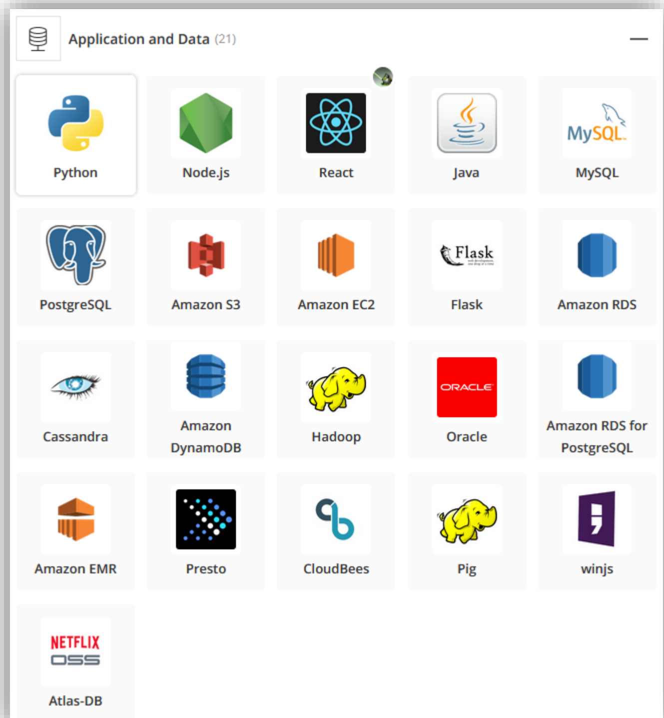
- Spotify

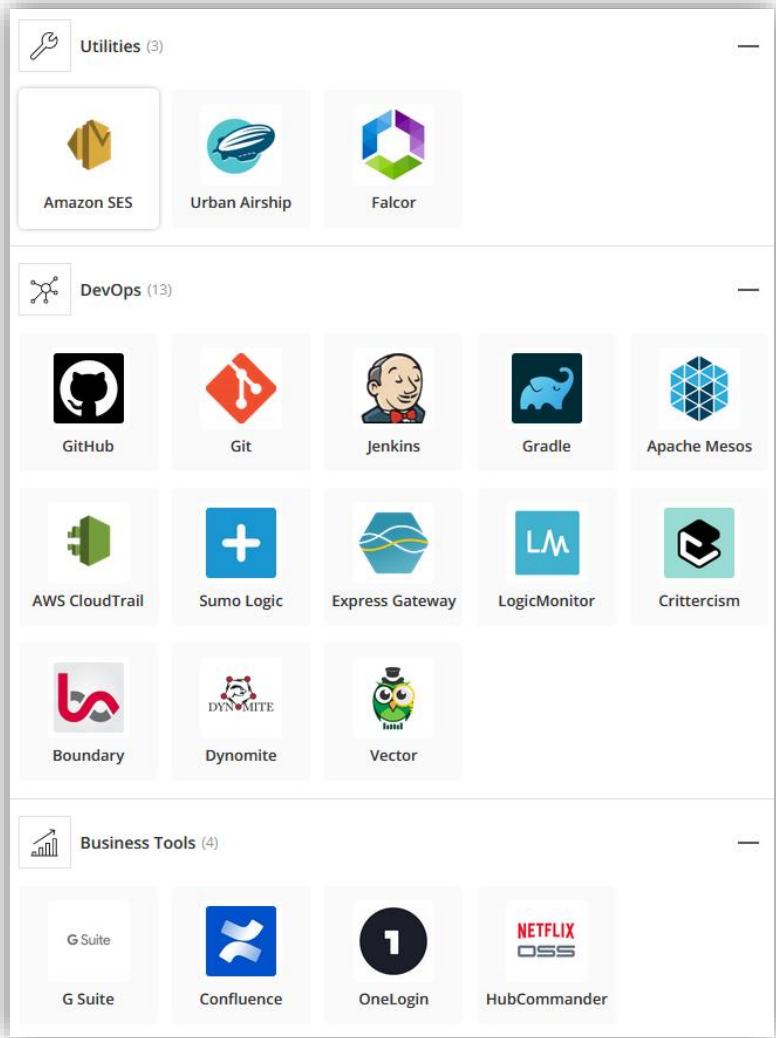




Technology Stack

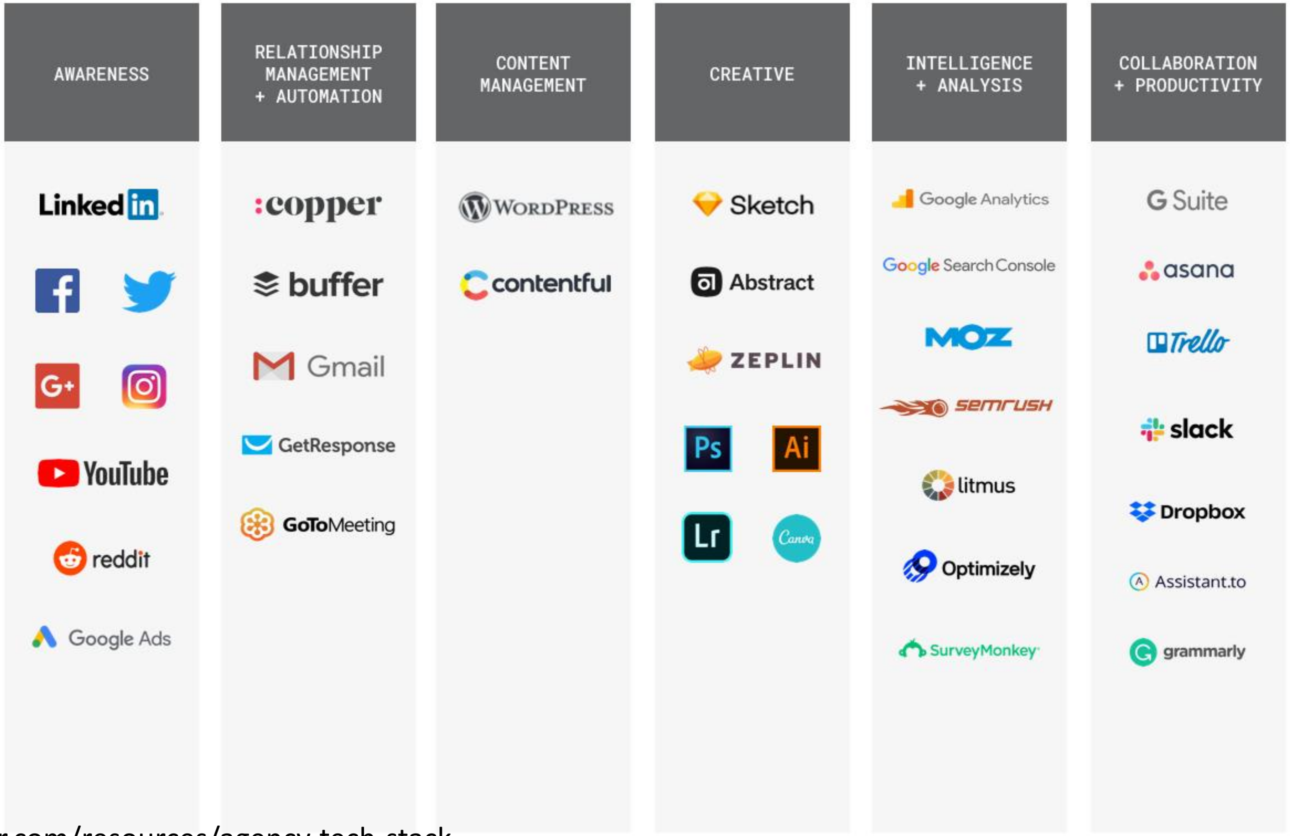
- Netflix





Technology Stack

- Tech stack in other context (e.g. marketing and creative agency)



Credit: https://www.copper.com/resources/agency-tech-stack

INTERNET & WWW

The Internet

- Internet is a network of interconnected computers that is now global
- Internet born in 1969 - called ARPANET
- 1969 ARPANET was connection of computers at UCLA, Stanford, UCSB, Univ. of Utah

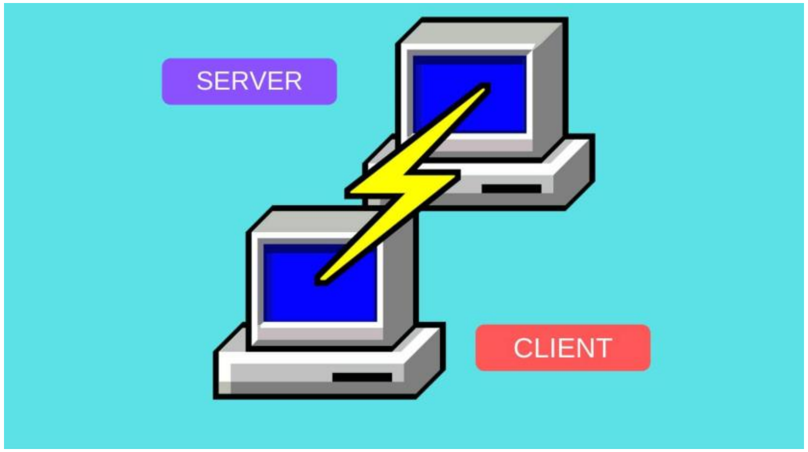
Computers late 60s & 70s

- No Personal Computers – all large mainframe computers in late 60s
- Mid 1970s – initial personal computers
  - Altair: Box with blinking lights
- Late 1970s – Apple 2, first usable PC

Credit: https://gamingdose.com/feature/altair-8800-จุดกำเนิดเล็ก-ๆ-ที่ยิ่/amp/
Credit: https://en.wikipedia.org/wiki/Apple_II

Internet 1970s

- 1972 - Telnet developed as a way to connect to remote computer
- 1972 – Email introduced
  - 1977 - U. Wisconsin has first “large” Email system - 100 users
- 1973 - ARPANET goes international
- 1973 - File Transfer Protocol (FTP) established



Credit: https://www.anet.net.th/a/43391

Computer 1980s

- 1981 – IBM PC
- 1984 – Apple Macintosh
- 1986 – Modem becomes option on PCs

Credit: https://th.wikipedia.org/wiki/ไอบีเอ็มพีซี
Credit: https://www.macthai.com/2014/01/24/30-years-of-mac-special-page-on-apple-website/

Internet 1980s

- 1984 - Domain Name Server introduced
  - allows naming of hosts, no longer numeric
- 1986 - NSFNET created
  - in 1990, becomes backbone of modern Internet when ARPANET is decommissioned
  - Completely privatized by 1995
  - 56 K interconnection initially, increased rapidly

Internet 1990s

- 1991 - Tim Berners-Lee releases **World Wide Web**!
  - TBL is computer programmer at CERN, a physics lab in Europe (book *Weaving the Web* by TBL)
- 1993 - Mosaic (becomes Netscape) designed by graduate students at University of Illinois
  - first point-and-click browser
  - later developed into Netscape Navigator
- These are the two most significant events in the formation of the WWW

Credit: https://www.anet.net.th/a/43391

World Wide Web

- Via Internet, computers can contact each other
- Public files on computers can be read by remote user
  - usually HyperText Markup Language (.html)
- HTTP - HyperText Transfer Protocol
- URL - Universal Resource Locator - is name of file on a remote computer

HTTP

- HyperText Transfer Protocol
- World Wide Web uses HTTP Servers, better known as web server
- Receive HTTP type request and send requested file in packets

Web Browsers

- Web browsers are the software we use to view web pages

URL

- Uniform Resource Locator
- URL is a reference to a resource that specifies the location of the resource on a computer network and a mechanism for retrieving it.
- URLs occur most commonly to reference web pages (http) and other related file like css, javascript, images, video, pdf, etc.
- Most web browsers display the URL of a web page above the page in an address bar.

URL Syntax

http://www.msu.edu/~urquhar5/tour/active.html

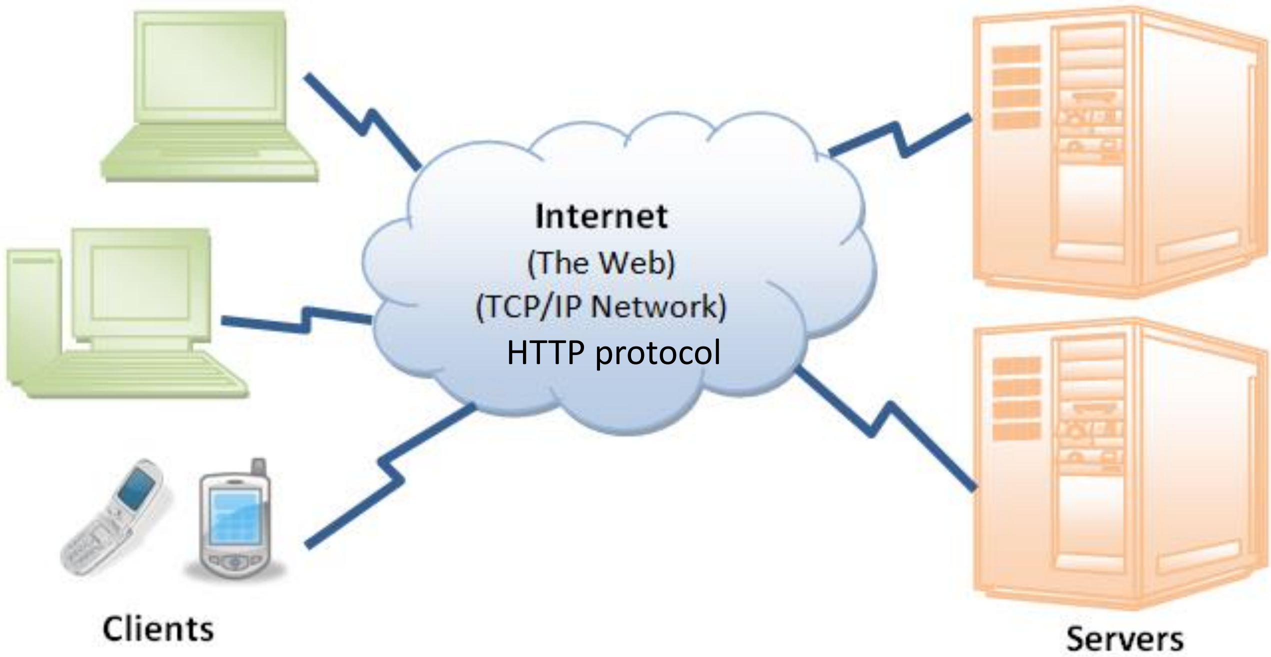
http://
identifies type
of transfer

/~urquhar5/tour/active.html
File Location on Remote Computer

www.msu.edu
Domain Name -
name of remote computer

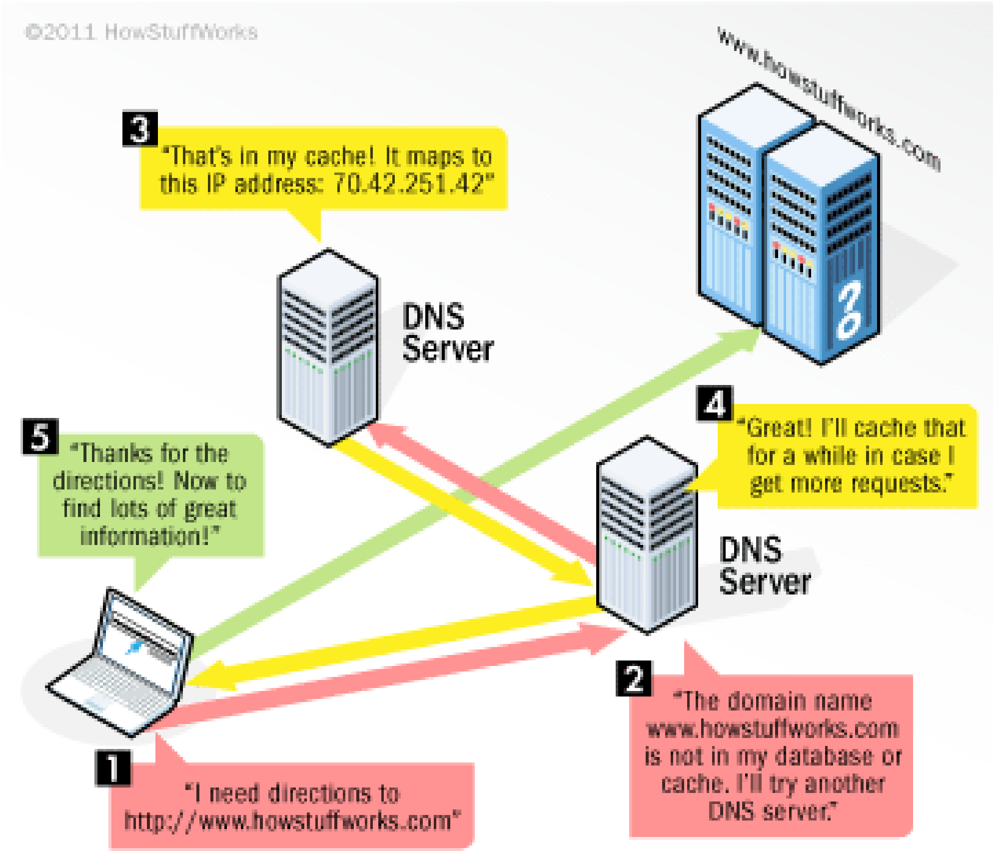
Credit: https://msu.edu/course/lbs/126/lectures/Internet-history.ppt

The WEB Architecture



Credit: https://www.ntu.edu.sg/home/ehchua/programming/webprogramming/HTTP_Basics.html

The WEB Architecture



Credit: https://commons.wikimedia.org/wiki/File:Dns-rev-1.gif

Website

- A website, also written as web site, or simply site, is a set of related web pages typically served from a single web domain.

- A website is hosted on at least one web server, accessible via a network such as the Internet or a private local area network through an Internet address known as a uniform resource locator (URL).

Webpage

- The building blocks of websites, documents, written in Hypertext Markup Language (HTML, XHTML).
- They may incorporate elements from other websites with suitable markup anchors.
- Accessed and transported with HTTP, HTTPS

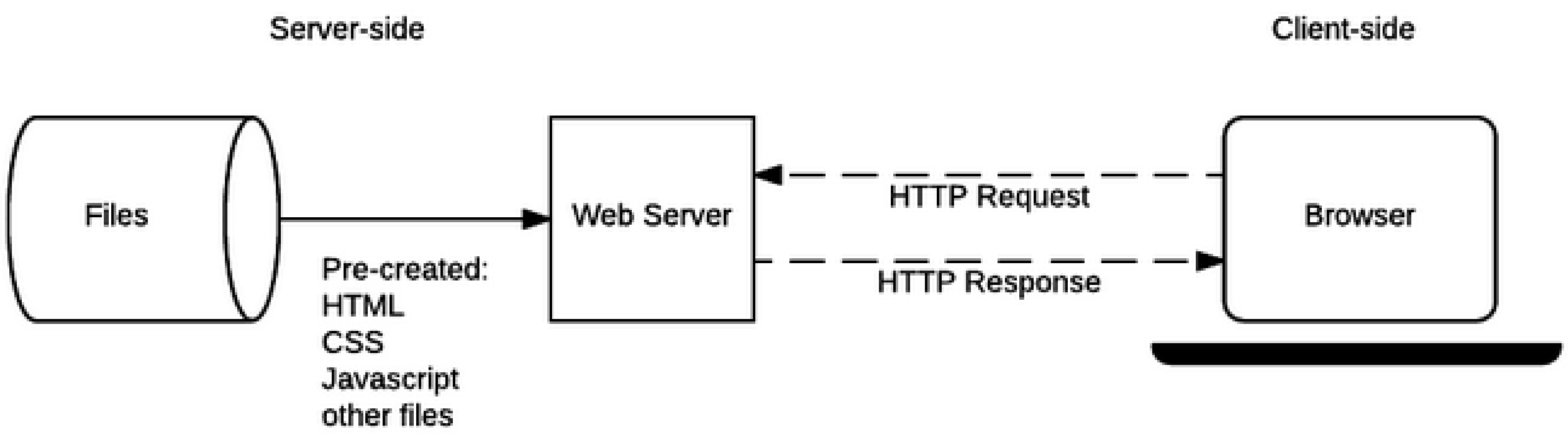
Static/Dynamic

- A static web page is delivered exactly as stored, as web content in the web server's file system

- A dynamic web page is generated by a web application that is driven by server-side software or client-side scripting. Dynamic website pages help the browser (the client) to enhance the web page through user input to the server.

The WEB Architecture

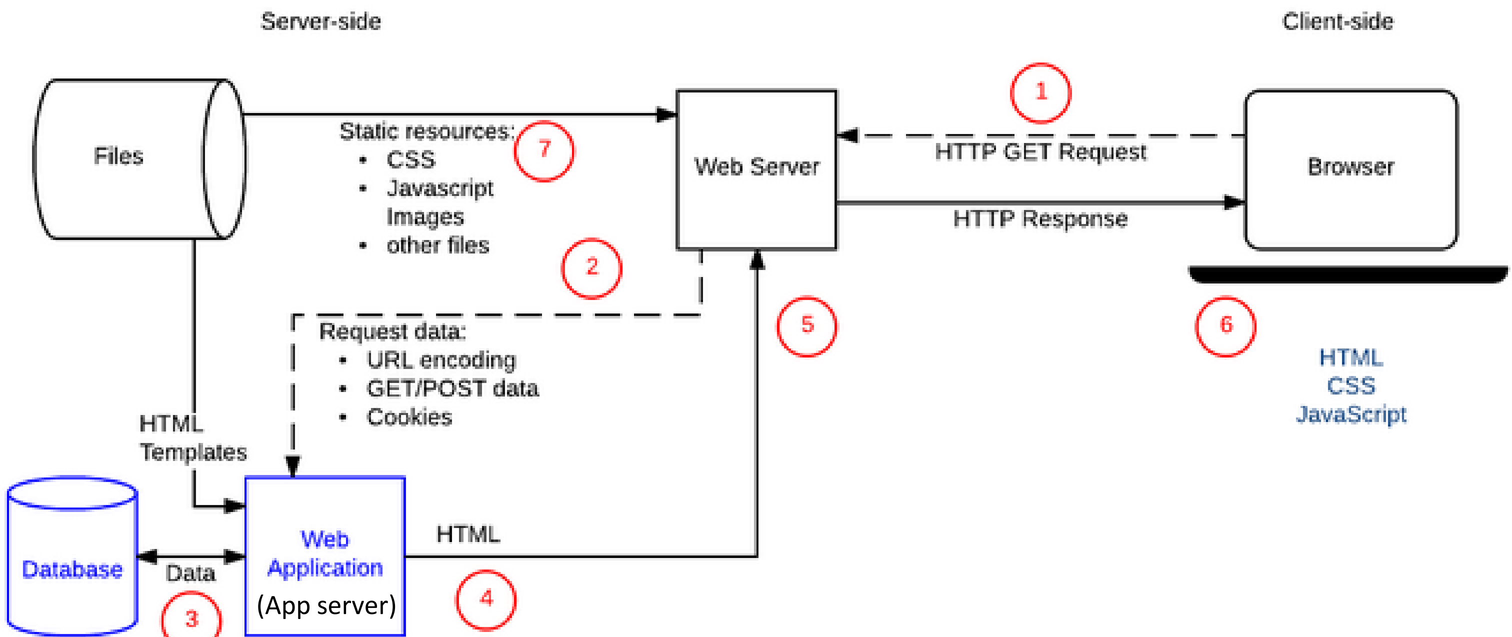
- Static sites



Credit: https://developer.mozilla.org/en-US/docs/Learn/Server-side/First_steps/Introduction

The WEB Architecture

- Dynamic sites



- **MySQL, Oracle, MS Access, etc. (SQL)**
- **MongoDB (NoSQL)**

- **JavaEE**
- **PhP**
- **nodeJS**

Credit: https://developer.mozilla.org/en-US/docs/Learn/Server-side/First_steps/Introduction

Web Address

- Identifier of email is email address like: peter.cop@somemail.com

- Identifier of web page is URL. The pages of a website can usually be accessed from a simple Uniform Resource Locator (URL) called the web address.

- The uniform resource locator (URL) of a home page is most often of the form http://domain.tld/index.htm or http://domain.tld/default.htm.

What Inside Webpage

- Perceived (rendered) information:
  - Textual information: with diverse render variations.
  - Non-textual information:
    - Static images: GIF, JPEG or PNG, SVG
    - Animated images: Animated GIF and PNG
    - Dynamic animated images, Flash, or Java applet
    - Audio: MP3, ogg format
    - Video: mpeg4, WMV (Windows), RM (RealMedia),
    - FLV (Flash Video), MPG, MOV (QuickTime)